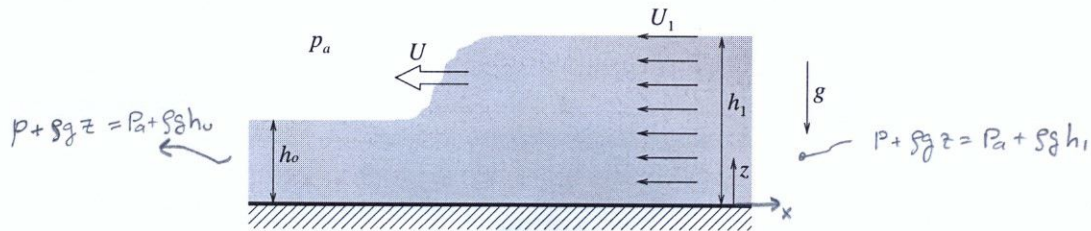
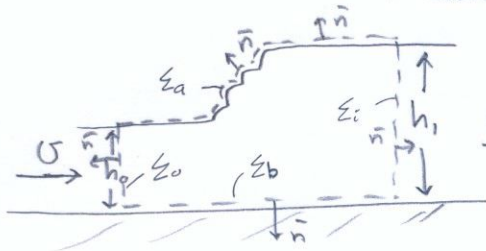


The bore sketched in the figure moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0 , leaving behind a layer thickness h_1 moving with velocity U_1 . Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U , h_0 and g . As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.



THE PROBLEM SIMPLIFIES IF DESCRIBED WITH A MOVING REFERENCE FRAME MOVING WITH VELOCITY $-U\bar{e}_x$



$$\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma_c} \rho (\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma$$

$$\int_{\Sigma_0 + \Sigma_i + \Sigma_b + \Sigma_a} \rho \bar{v} \cdot \bar{n} d\sigma = -\rho U h_0 + \rho (U - U_1) h_1 = 0 \Rightarrow U h_0 = (U - U_1) h_1$$

$$\frac{d}{dt} \int_{V_c} \rho \bar{v} dV + \int_{\Sigma_c} \rho \bar{v} [(\bar{v} - \bar{v}_c) \cdot \bar{n}] d\sigma = - \int_{\Sigma_c} p \bar{n} d\sigma + \int_{\Sigma_c} \bar{z}' \cdot \bar{n} d\sigma = \int_{V_c} \rho g \bar{e}_z dV$$

$$-\rho U U \bar{e}_x h_0 + \rho (U - U_1)^2 \bar{e}_x h_1$$

IT CAN BE IN GENERAL WRITTEN AS

$$- \int_{\Sigma_c} (p - p_a) \bar{n} d\sigma$$

$$+ \int_{\Sigma_b} (p - p_a) d\sigma \bar{e}_z + \int_0^{h_0} \rho g (h_0 - z) dz \bar{e}_x - \int_0^{h_1} \rho g (h_1 - z) dz \bar{e}_x$$

$$\int_{\Sigma_c'} \bar{z}' \cdot \bar{n} d\sigma = \int_{\Sigma_0 + \Sigma_i} \bar{z}' \cdot \bar{n} d\sigma + \int_{\Sigma_a} \bar{z}' \cdot \bar{n} d\sigma + \int_{\Sigma_b} \bar{z}' \cdot \bar{n} d\sigma$$

X-COMPONENT OF THE MOMENTUM EQUATION

$$-\rho U^2 h_0 + \rho (U - U_1)^2 h_1 = \frac{\rho g}{2} (h_0^2 - h_1^2)$$

$$-\rho U h_0 U_1 = \frac{\rho g}{2} (h_0^2 - h_1^2)$$

$$\frac{h_1}{h_0} = \frac{1}{2} (\sqrt{1 + 8F} - 1)$$

$$\frac{U_1}{U} = \frac{3}{2} \left(\frac{1}{3} \sqrt{1 + 8F} - 1 \right)$$

Z-COMPONENT OF THE MOMENTUM EQUATION

$$\int_{\Sigma_b} (p - p_a) d\sigma = \int_{V_c} \rho g dV$$

$$\text{with } F = \frac{U^2}{gh_0}$$